



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	CASE-df005-2017-ios
Operator:	Floris
Analysis Started:	2026-06-15 14:50:32
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\test_phantom_4\df005\2017\ios
Output Directory:	C:\Users\Floris\Documents\dji_drones\test_phantom_4\df005\2017\dfdof_output\CASE-df005-2017-ios
Evidence Sources:	2
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df005_external.001	drone_sd	physical	14.9 GB	2026-06-15 14:51:19
DF005_flight.zip	controller_ios	logical	263.1 MB	2026-06-15 14:51:20

Evidence Source Path

C:\Users\Floris\Documents\ddj_drones\test_phantom_4\df005\2017\ios

Cryptographic Hashes

df005_external.001		drone_sd	physical
SHA-256:	15f4b7897912ab70c151f23ff4050677e622b2d84412c7d6c6616952d9f8f7e8		
SHA-1:	dc26935a64954a24a41776cba813b4c1dea80017		

DF005_flight.zip		controller_ios	logical
SHA-256:	6533318645cf08b92aa8ebdbb7ab22980c428a76f9f9a8b252e349bd8988488d		
SHA-1:	0697a1a8fa08a3928a62c63fa3e33e8a7c0f48d5		

Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p3 - controller android]: source evidence not found
3. [p3 - drone flight storage]: source evidence not found
4. [p4 - controller android]: source evidence not found
5. [p4 - drone flight storage]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
datcon	1	1	[OK]
exiftool	31	31	[OK]
fls	1	1	[OK]
icat	25	25	[OK]
mmls	1	1	[OK]
parser_ios	1	1	[OK]
txtlogtocsvtool	1	1	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	2 / 4
Artefact Categories With Data	7 / 7
Flights With Primary Correlation	1 / 1
Tools Succeeded	7 / 7

3. Device and Account Identification

Field	Value	Confidence	Sources
Installed DJI App(s)	com.dji.go com.dji.pilot	[K] Controller-only	1
Controller Device	iPad mini 4	[K] Controller-only	1
Drone Serial Number	07JDE5300200GN	[K] Controller-only	1
Drone Model	P4 Professional P4P	[!] Inconsistent	2
Controller Device	iPad5,1__weibosdk__003143000__iphone__os 10.3.1 iPad5,1__weibosdk__003143000__iphone__os 10.3.2	[!] Inconsistent	2
DJI Application Version	3.1.12 4.1.3	[!] Inconsistent	3
Account E-Mail	droneforensics@vtoinc.com	[K] Controller-only	2
Product Type	1 24	[!] Inconsistent	2
Last Country Code	US	[K] Controller-only	1
Last App Launch	2017-06-28T19:40:33Z 2017-06-29T19:49:43Z	[!] Inconsistent	2
Last Flight Date	1970-01-01T00:00:00Z	[K] Controller-only	2

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status: Matched (primary GPS+temporal)
Confidence: high
GPS Overlap: 622.9 s
Median GPS Distance: 1.18 m
Overlap: 100%
GPS Match Rate: 100%
Flight Record SHA-256: 7396b1d790b29b98bb48a049a3d4e9cc7a18270ddba241d3786bc6bce199d8b8
Flight Record Name: DJIFlightRecord_2017-06-29__14-52-06_.csv
Start: 2017-06-29T20:52:06.536000+00:00
End: 2017-06-29T21:02:29.462000+00:00
Duration: 622.9 s
Drone Log SHA-256: 59ff6259218b8573578840fb71f6782bc692b6b102c606e2cd0d2cdf5671aa4a
Drone Log Name: 2017-06-29_14-51-13_FLY005.csv
Start: 2017-06-29T20:51:01.985000+00:00
End: 2017-06-29T21:02:55.986000+00:00
Duration: 714.0 s

Flight Metrics

Peak Height: 47.7 m
Distance Travelled: 2319.61 m
Video Recording: Yes
Photo Capture: Yes
Photos Taken: 7
Recording Duration: 255.0 s
Record Complete: No - drone had not returned to home point

Fly Modes Observed (Chronological)

1. Atti
2. GPS_Atti
3. ASST_TAKEOFF
4. EngineStart
5. AutoTakeoff

Key Events

Timestamp (UTC)	Source	Event	Details
-	ctrl_ios:dl	Motor turned off	off
2017-06-29 20:51:31	ctrl_ios:dl	Log started	P4P
2017-06-29 20:52:06	ctrl_ios:dl	Motor turned on	on
2017-06-29 20:52:06	ctrl_ios:fr (v4.1.3)	Log started	SN: 07JDE5300200GN; App: 4.1.3; P4 Professional
2017-06-29 20:52:06	ctrl_ios:fr (v4.1.3)	Motor turned on	on
2017-06-29 20:52:07	ctrl_ios:fr (v4.1.3)	Record mode changed	No
2017-06-29 20:53:11	ctrl_ios:fr (v4.1.3)	Record mode changed	Start
2017-06-29 20:53:12	ctrl_ios:fr (v4.1.3)	Record mode changed	Starting
2017-06-29 20:56:52	ctrl_ios:fr (v4.1.3)	Record mode changed	Stop
2017-06-29 20:56:54	ctrl_ios:fr (v4.1.3)	Record mode changed	No
2017-06-29 20:56:55	ctrl_ios:fr (v4.1.3)	Record mode changed	Starting
2017-06-29 20:57:31	ctrl_ios:fr (v4.1.3)	Record mode changed	Stop
2017-06-29 20:57:32	ctrl_ios:fr (v4.1.3)	Record mode changed	No
2017-06-29 21:00:18	ctrl_ios:fr (v4.1.3)	Reached peak height	47.7 m
2017-06-29 21:00:57	ctrl_ios:dl	Reached peak height	37.5 m

Timestamp (UTC)	Source	Event	Details
2017-06-29 21:02:29	ctrl_ios:dl	Motor turned off	off
2017-06-29 21:02:29	ctrl_ios:fr (v4.1.3)	Log ended	Dist home: 1.5 m; Height: 7.0 m; Rec: 255.0 s
2017-06-29 21:02:55	ctrl_ios:dl	Log ended	
2017-06-29 20:53:55	ctrl_ios:fr (v4.1.3)	Photo mode changed	Single
2017-06-29 20:53:59	ctrl_ios:fr (v4.1.3)	Photo mode changed	Single
2017-06-29 20:54:08	ctrl_ios:fr (v4.1.3)	Photo mode changed	Single
2017-06-29 20:56:35	ctrl_ios:fr (v4.1.3)	Photo mode changed	Single
2017-06-29 20:56:39	ctrl_ios:fr (v4.1.3)	Photo mode changed	Single
2017-06-29 20:56:58	ctrl_ios:fr (v4.1.3)	Photo mode changed	Single
2017-06-29 20:57:05	ctrl_ios:fr (v4.1.3)	Photo mode changed	Single

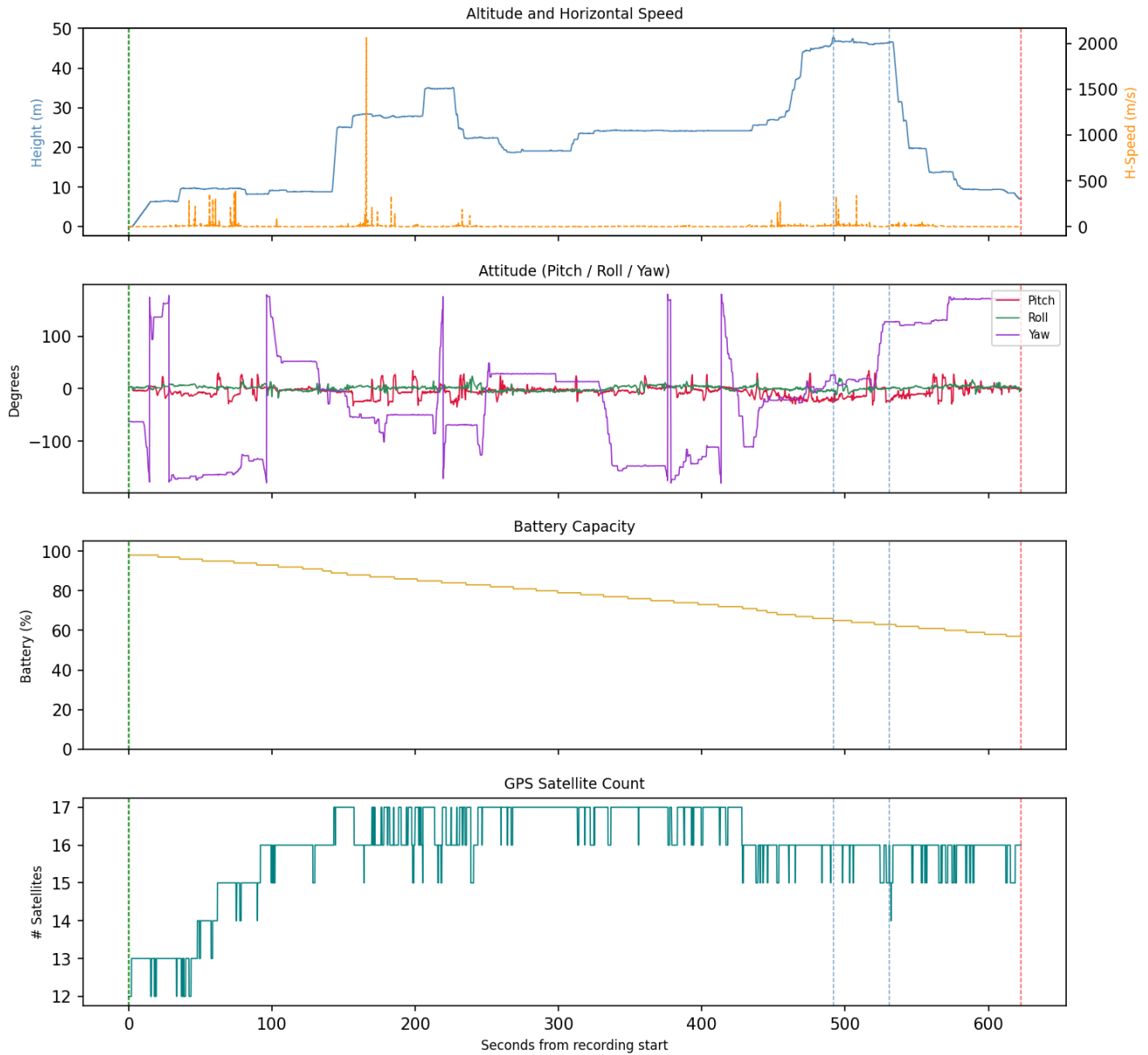
Correlated Artefacts

Possibly correlated (indirect indicators):

drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:7b8d41c1775714af2eadff7cf56c9acbf4b0d80a1c53867d061be7e25eea680	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:83d34e5880b8be3bde56bd87afebd103592a5e5994304a995d036034962a8272	
drone_sd:videos	GPS within flight bounding box (lat 39.9585, lon -106.2203)
p5:84b82a66fa36d126ed580f2d91bc47d6f4787b93bec75585ba23dfef4068eb19	
drone_sd:videos	GPS within flight bounding box (lat 39.9585, lon -106.2203)
p5:624ccfe3d1f63aad44f49156a8f89326e679a81087999ebb77f6707815e350bb	
drone_sd:videos	GPS within flight bounding box (lat 39.9611, lon -106.2166)
p5:07193b3b3f1f024c7b8e66cbe104bd4106db3269a6f278245056cb63ec837132	
drone_sd:videos	GPS within flight bounding box (lat 39.9611, lon -106.2166)
p5:d1c7d4c9532dd338776e728447b3f9ac9637aa0df2403597cebae87a12fce7ae	
drone_sd:videos	GPS within flight bounding box (lat 39.9582, lon -106.2172)
p5:c29a9498a07292ad1a133d77584e928ee5a05147cfd4d3b770f4d7c6ada17406	
drone_sd:videos	GPS within flight bounding box (lat 39.9582, lon -106.2172)
p5:14fbc8239759ab3953aca18c3b7ef9beccc96ee09c67943bfe7483dbfff99068	
drone_sd:videos	GPS within flight bounding box (lat 39.9588, lon -106.2202)
p5:7eecd89bd8c561442c19cd3b6c462e508398bb945657f0acb4fb3a3b29b18f55	
drone_sd:videos	GPS within flight bounding box (lat 39.9588, lon -106.2202)
p5:bb2ec42222e8a87726519e1846155bc5937aa0b31f6a797cd97e71376e65aa45	

Flight Record - Sensor Telemetry

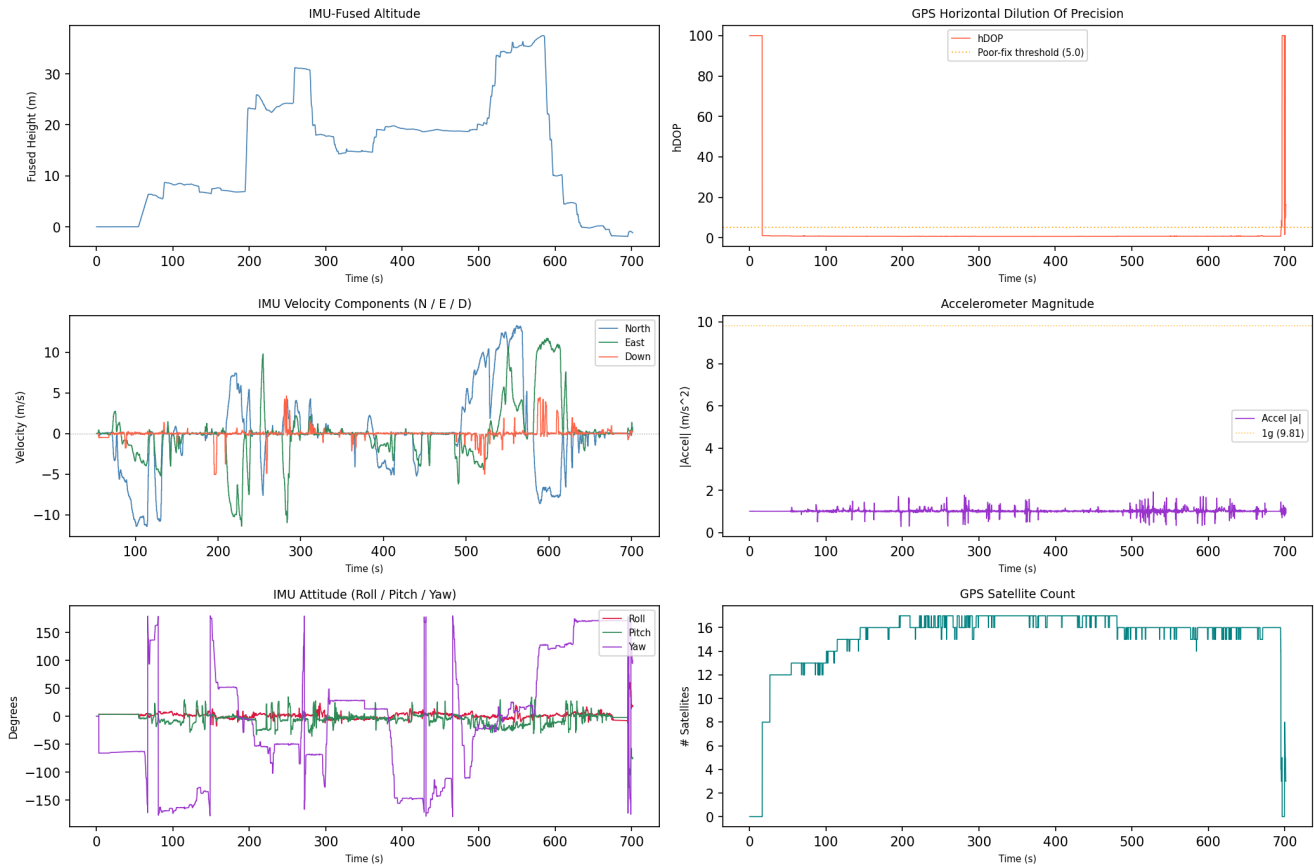
Flight Record Telemetry - flight_01



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

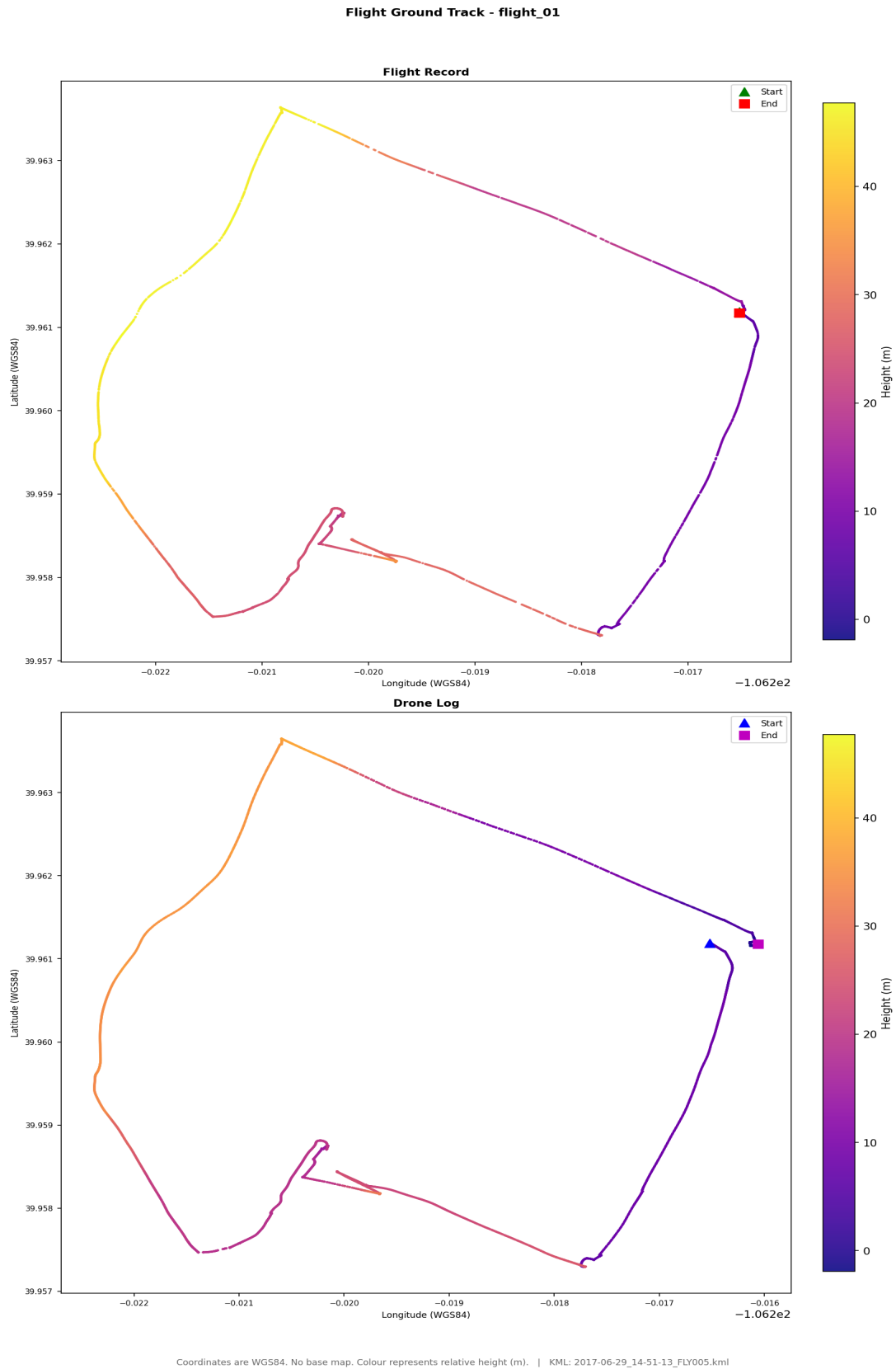
Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_01



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (2017-06-29_14-51-13_FLY005.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path.

Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	No
controller_ios	Yes
drone_sd	Yes
drone_flight_storage	No

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_ios	2	12	1	3	1	3	3
drone_sd	0	0	0	0	0	15	10

Data Quality Notes per Category

Databases (12 files):

- 4 database(s) empty (controller_ios)

Drone Logs (1 files):

- 1 file(s) with row anomalies
- 1 file(s) with empty columns
- 1 file(s) with constant-value columns

Flight Logs (3 files):

- 1 file(s) unreadable (binary)
- formats: 1 bracketed_log, 1 json_array_log

Flight Records (1 files):

- 1 file(s) with empty columns
- 1 file(s) with constant-value columns

Images (18 files):

- 15 file(s) contain(s) EXIF GPS coordinates
- 18 file(s) missing (normalisable) EXIF timestamp
- 3 file(s) missing EXIF GPS coordinates

Videos (13 files):

- 10 file(s) contain(s) EXIF GPS coordinates
- 13 file(s) missing (normalisable) EXIF timestamp
- 3 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Drone_logs: 1 file(s) with row anomalies detected in telemetry data.
- Drone_logs: 1 file(s) with empty columns - some telemetry channels contain no data.
- Drone_logs: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.
- Flight_logs: 1 file(s) unreadable (binary) - file content could not be parsed.
- Flight_logs: log files present in the following format(s): 1 bracketed_log, 1 json_array_log.
- Flight_records: 1 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p3 - controller android]: source evidence not found
3. [p3 - drone flight storage]: source evidence not found
4. [p4 - controller android]: source evidence not found
5. [p4 - drone flight storage]: source evidence not found

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Databases

djiFMDB.db

- database_empty: all tables empty

djiFMDB_1.db

- database_empty: all tables empty

statistics.db

- database_empty: all tables empty

statistics_1.db

- database_empty: all tables empty

Drone Logs

norm_2017-06-29_14-51-13_FLY005.csv

- missing_gps: 4 row(s) affected
- zero_coordinate: 285 row(s) affected
- altitude_negative: 839 row(s) affected
- gps_poor_fix: 348 row(s) affected
- contains_no_value: 4 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C
- contains_constant_value: 37 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, BatteryInfo:dyna_cnt:D, BatteryInfo:f_cap:D, BatteryInfo:out_ctl:D (+32 more)
- gps_accuracy_poor: 361 row(s) affected
- quaternion_invalid: 51 row(s) affected
- attitude_out_of_bounds: 49 row(s) affected

Flight Logs

2017-06-29_14_52_06.dat

- format: json_array_log | entries: 4

upgradeLog_0.txt

- format: bracketed_log | entries: 21

Flight Records

norm_DJIFlightRecord_2017-06-29_14-52-06_.csv

- contains_no_value: 29 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_WARN.warn,

CENTER_BATTERY.isBatteryOnCharge (+24 more)

- contains_constant_value: 171 column(s) - APP_SER_WARN.warn, CAMERA_INFO.cameraType, CAMERA_INFO.connectState, CAMERA_INFO.encryptStatus, CAMERA_INFO.firmErrorType (+166 more)

Images

2017_06_29_14_53_12.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_14_56_54.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_15_02_05.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0003.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0004.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0006.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0007.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0008.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0010.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0011.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0013.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0014.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0015.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0016.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0017.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0019.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0020.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

Videos

2017_06_29_14_53_12.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_14_56_54.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_15_02_05.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0001.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0005_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0005.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0009_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0009.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0012_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0012.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0018_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0018.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Drone Model inconsistency detected - conflicting values observed: P4 Professional, P4P. Manual verification required.
2. Controller Device inconsistency detected - conflicting values observed: iPad5,1__weibosdk__003143000__iphone__os10.3.1, iPad5,1__weibosdk__003143000__iphone__os10.3.2. Manual verification required.
3. DJI Application Version inconsistency detected - conflicting values observed: 3.1.12, 4.1.3. Manual verification required.
4. Product Type inconsistency detected - conflicting values observed: 1, 24. Manual verification required.
5. Note: no firmware version found - Device and account information.
6. Last App Launch inconsistency detected - conflicting values observed: 2017-06-28T19:40:33Z, 2017-06-29T19:49:43Z. Manual verification required.
7. Note: no last known latitude (find aircraft) found - Device and account information.
8. Note: no last known longitude (find aircraft) found - Device and account information.
9. Note: no last known altitude (find aircraft) found - Device and account information.
10. Note: no last known heading (find aircraft) found - Device and account information.
11. Note: no last known horiz. accuracy (find aircraft) found - Device and account information.
12. Flight record(s) show signs of incomplete capture (flight_01). The drone may not have returned to the home point before logging stopped. Manual inspection recommended.
13. Photo(s) were recorded during the flight. Copies may reside on both the controller and the drone SD card - both sources should be examined.
14. 8 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

com.dji.go.plist		Account Data	controller_ios	extract_logical
SHA-256:	e67089356bdfa672298845e870fce5a6bc313ed6b7a59b8ee70643376dd9da2e			
SHA-1:	2be29a310d8db94e1cb6c815c9ff4f9ee55819b5			

com.dji.pilot.plist		Account Data	controller_ios	extract_logical
SHA-256:	4f4171f18adb39e29b892641d9f90fcfd1d20d8c746e336df97f5fa679c8c94d7			
SHA-1:	f20f2831618174944640591a9b4ecef51a2ac540			

airmap_flysafeplaces.db		Databases	controller_ios	extract_logical
SHA-256:	510c575a7fb8deb6362753c3b6986c8d5fb7b56d93929bc7f3551a31a631a0cd			
SHA-1:	45d26e1d840cb34c497d2bf4af67a5797163a442			

nfz_flysafeplaces.db		Databases	controller_ios	extract_logical
SHA-256:	01e11745b0203c510fc960c236e8a4244f84149493485a98a94c25b0627817b2			
SHA-1:	4933b833a2709e5819c1eea49fecc4c673944932			

statistics.db		Databases	controller_ios	extract_logical
SHA-256:	83bfc672a35016dcdfa168cbc780d6219345ddfe601ad75087ce879503a3058a			
SHA-1:	3a8843449db0f78d4502b81d67e0f26584226a80			

tencent_analysis.db		Databases	controller_ios	extract_logical
SHA-256:	6aa861666d5c1cefb39177a4f250acf0c4b9f4221d8b023f67b7b14c530293d0			
SHA-1:	e9fc874b10c755a97be0c8ff523be486b39681a8			

tencent_wxo_analysis.db		Databases	controller_ios	extract_logical
SHA-256:	9ce708a5c4f2b2a92448f0f54dcb7c0176da54656a7b26ab0d7937ed4f4bb7b8			
SHA-1:	4d74aa8bc60ad4fbc4a4b863cd8bc8c6218d186			

djiFMDB.db		Databases	controller_ios	extract_logical
SHA-256:	baa7277998f1c5db44be2b44a2264bedd94f93495fbb0d73636ee2fdc9c94848			
SHA-1:	d6720f20112c173533a7ad8ccedd8c8992503ddb			

airmap_flysafeplaces_1.db		Databases	controller_ios	extract_logical
SHA-256:	01472f913caf3ce5d50b146cd37a6fea1a18ca4d9666b2d5dc06189a8ac5c17c			
SHA-1:	fdecfca00525fd0f36dc51b5a8709efaf409f908			

nfz_flysafeplaces_1.db		Databases	controller_ios	extract_logical
SHA-256:	a28ac9537a932b4f268c4c3e1602bad52ab295a31a92ced957a812fbfc2a154a			
SHA-1:	57e091e1450ad987218fd579127057038386ee9e			

statistics_1.db		Databases	controller_ios	extract_logical
SHA-256:	83bfc672a35016dcdfa168cbc780d6219345ddfe601ad75087ce879503a3058a			
SHA-1:	3a8843449db0f78d4502b81d67e0f26584226a80			

tencent_analysis_1.db		Databases	controller_ios	extract_logical
SHA-256:	b760984b56b9c85b4fb9f3c37e1d01bbcb2e5c9ec8b990693750b08ca987fd60			
SHA-1:	c7cfa530c25ae3012b87409158f4cf47bd53b09b			

tencent_wxo_analysis_1.db		Databases	controller_ios	extract_logical
SHA-256:	da99379fed69224222da353ef73568d11b1fbb80dd3d845848d5f96a0afb5746			
SHA-1:	cb18325b8ba7a8377b7f9da190b9e3fa4c311cfa			

djiFMDB_1.db		Databases	controller_ios	extract_logical
SHA-256:	003687bc29cf9f101fae000e55fef1718de111f65242f517487ab2cb826a3e37			
SHA-1:	86fe40031cfeab9bdb5627a6b0bb7a446abb05f1			

2017-06-29_14-51-13_FLY005.DAT		Drone Logs	controller_ios	extract_logical
SHA-256:	942292720321e06884699a1d3095f0908981e7f408661f17bdc57ecc7f5d92c7			
SHA-1:	c9f9a067619effc8e894c0417975c99e4c22c43e			

app_log.txt		Flight Logs	controller_ios	extract_logical
SHA-256:	0fc709740594b6de6daa40f0de7966720fab7ffab5465462b35cb4a73a1cbd5a			
SHA-1:	90bff1d9c60278a9613e82005f5edaae5e4e3cd4			

2017-06-29_14_52_06.dat		Flight Logs	controller_ios	extract_logical
SHA-256:	7b69bb1e060c2fc6bdf1bede25b5ad35b298a2b19e14611406cb7d2f8b6dcf64			
SHA-1:	08446e08ecf1bdf8d967cbf5ff50765762abf848			

upgradeLog_0.txt		Flight Logs	controller_ios	extract_logical
SHA-256:	fde8552af3146c8354ad025ba34986b37879032a8aee20e568ab2d7637dfa124			
SHA-1:	1c5349f1a47c6717fd0a03dde55fbc8c54c73b7e			

DJIFlightRecord_2017-06-29_14-52-06_.txt		Flight Records	controller_ios	extract_logical
SHA-256:	168634ecd25fd9d4872fbd0d454a8338976c1e0500f764d9d282916fb7b447e9			
SHA-1:	45c98e0ff353dafd553e20f186c2574c422fe9ab			

2017_06_29_14_53_12.thumbnail		Images	controller_ios	extract_logical
SHA-256:	0abc5b81ed682110ec623a99000343a07879a463f900d88f05e95771ad96b59c			
SHA-1:	d3fe0d732bc2ff1064e50b11650ce70aff01cda4			

2017_06_29_14_56_54.thumbnail		Images	controller_ios	extract_logical
SHA-256:	f55abc672a911d4c60fcf5ce199aa19ecae00a4ef2ea39830dde133894ea3ced			
SHA-1:	014cb28e38a8d6b0d098e2079726294094a27bca			

2017_06_29_15_02_05.thumbnail		Images	controller_ios	extract_logical
SHA-256:	1c873073e7571a7328ce66f7025b7ab0b3d0f1330c5f2fa594c97111d606ab63			
SHA-1:	42d03f3512abd1717404e4d20831428d9fdcbe00			

2017_06_29_14_53_12.mp4		Videos	controller_ios	extract_logical
SHA-256:	76339b5695a7456a5f6efae11ee45589d90e671e9631812b704ddab476b569f5			
SHA-1:	c9384908d05eb641fb4e53d18fd0b6367619505d			

2017_06_29_14_56_54.mp4		Videos	controller_ios	extract_logical
SHA-256:	75c4640ac54dbbcf03ae840be683e37ee59eac9c767ec5fff603fe5cfdab159d			
SHA-1:	9c89915ee95e9841437d6665e772bf18935eb879			

2017_06_29_15_02_05.mp4		Videos	controller_ios	extract_logical
SHA-256:	2cb5e19c1a312684497ec8bef3a559f451ff593cc960185104ceb4d91702d96a			
SHA-1:	992b841180e7688c9c2fbff769fce0eea7987bb3			

DJI_0002.JPG	Images	drone_sd	extract_physical
SHA-256:	abc58ba30b10e8280e07a11db5163a9874290dd76bfb8124d8c8178bcac72124		
SHA-1:	dce2738dfd79c8c9c9bd1e3f68257eeac72f6a7		

DJI_0003.JPG	Images	drone_sd	extract_physical
SHA-256:	a80595fc4dc562dda8355ea5645ba27de8c29cebe7f4b860a9f022c94a878451		
SHA-1:	91ae86314d46b9a814df5486995b46d99fc8e3e3		

DJI_0004.JPG	Images	drone_sd	extract_physical
SHA-256:	d9ad93a5a6b0d831685e27ad41c9342b7e7217198152d851d3e331ee19c510f3		
SHA-1:	b2d0f4f11d687f2dd867c6be5896b2e4acd78c37		

DJI_0006.JPG	Images	drone_sd	extract_physical
SHA-256:	744ab3c8b893b3fdb40cbe3c6ec70756a2f6f78d6dfc94612bd66e9f809f79fa		
SHA-1:	d1770f3dcc6cd2a28161a48a2feb8a911c4baf44		

DJI_0007.JPG	Images	drone_sd	extract_physical
SHA-256:	fe35c4c350c39e33b95fbb8034dc13a7c9f6cc7eb169e451e85499e27251d479		
SHA-1:	74661fca261a4e74b5f7c379ce135af2b2db12b9		

DJI_0008.JPG	Images	drone_sd	extract_physical
SHA-256:	8478e34516836246cdca8cee8bfd13c82ae95b72aa26e80e9c12c4de900fff31		
SHA-1:	7afe816f41819d5508891e68ea35923f4fa20fce		

DJI_0010.JPG	Images	drone_sd	extract_physical
SHA-256:	3148d9fb0072c4522a14bff5cc8a6d8e35850852d56b97114de83cf349b3365a		
SHA-1:	ae8bd84e6b219ffe416a891a17ad4d6ca298a2d0		

DJI_0011.JPG	Images	drone_sd	extract_physical
SHA-256:	2d076163a3d537969bcc9d34fb421a6930061eaf2b1c883eb24c6ad310fe2dd1		
SHA-1:	2e8a9126c020c853990f26fc9cd46ee192682b1b		

DJI_0013.JPG	Images	drone_sd	extract_physical
SHA-256:	6f7b2a40733e00f1c52772fa209dec54350a1b9097dac28d13179a5992bb43b3		
SHA-1:	f443af9bb57ab2d76e8cf8f3a99cb93ea85ec605		

DJI_0014.JPG	Images	drone_sd	extract_physical
SHA-256:	6e5b681afab934878096c909454bf4a257b01d3293bf69abb8adfb48053e74a9		
SHA-1:	f0fc2cb6a796ba55199db3f3a15fa3296b1d51e3		

DJI_0015.JPG	Images	drone_sd	extract_physical
SHA-256:	79d467f059826c93617dc94a8ad5ed32234c3d75adf043cc0cc1868ccc4c8f09		
SHA-1:	6d3f148abc9f1926347c387e9a66346aa4a14c77		

DJI_0016.JPG	Images	drone_sd	extract_physical
SHA-256:	a6c877233493fd603062ed737615fee8a8ceb55ba2b6da09e5b19a232e0b76d9		
SHA-1:	a1ab9fc323ef5a77870c1b5bea8df1c72fffc9344		

DJI_0017.JPG	Images	drone_sd	extract_physical
SHA-256:	eee797d80b320a4328022be386c0d9c825c3eb18d86292a069abac1e7da14b58		
SHA-1:	972d904cd04e16d6392df4f5585d696744d544f7		

DJI_0019.JPG		Images	drone_sd	extract_physical
SHA-256:	5cd0b44634eb69338cc27858c94f286628ff58784f59ff3dc2307adec903c8ed			
SHA-1:	4238e64fa022c2e0a91a4c2fcd8438d00d907fac			

DJI_0020.JPG		Images	drone_sd	extract_physical
SHA-256:	7df85c310d22165acadfb8c9ce832761b368383f34ea42dae50bbcd011c8a2ea			
SHA-1:	1d42c6d0b52da2eed98c55b49fbb8205406eda63			

DJI_0001.MOV		Videos	drone_sd	extract_physical
SHA-256:	83d34e5880b8be3bde56bd87afebd103592a5e5994304a995d036034962a8272			
SHA-1:	19fb83d04583cf8ecddac75ca45229900afe9bb7			

DJI_0005.MOV		Videos	drone_sd	extract_physical
SHA-256:	624ccfe3d1f63aad44f49156a8f89326e679a81087999ebb77f6707815e350bb			
SHA-1:	8d7da3ee328562929d5a657b0b7292f349e84a90			

DJI_0009.MOV		Videos	drone_sd	extract_physical
SHA-256:	d1c7d4c9532dd338776e728447b3f9ac9637aa0df2403597cebae87a12fce7ae			
SHA-1:	6083d7e5f675e1f216ef82fa9814f4411e40634f			

DJI_0012.MOV		Videos	drone_sd	extract_physical
SHA-256:	14fbc8239759ab3953aca18c3b7ef9beccc96ee09c67943bfe7483dbfff99068			
SHA-1:	a54f6de5eef35eba6ca50cc376aefb3190d068b2			

DJI_0018.MOV		Videos	drone_sd	extract_physical
SHA-256:	bb2ec4222e8a87726519e1846155bc5937aa0b31f6a797cd97e71376e65aa45			
SHA-1:	8d1f7ae484a6b6821af19b52ede60e7df313f403			

DJI_0001_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	7b8d41c1775714af2eadff7cf56c9acfbf4b0d80a1c53867d061be7e25eea680			
SHA-1:	605ca2b769e5b9892d7bf41b4b89f01e4e3c855d			

DJI_0005_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	84b82a66fa36d126ed580f2d91bc47d6f4787b93bec75585ba23dfeb4068eb19			
SHA-1:	8e9da7f5e2f24d861c5a46845cbf0b03aa4b4a0c			

DJI_0009_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	07193b3b3f1f024c7b8e66cbe104bd4106db3269a6f278245056cb63ec837132			
SHA-1:	e78d17014597433209d68326585d6ec003af7818			

DJI_0012_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	c29a9498a07292ad1a133d77584e928ee5a05147cfd4d3b770f4d7c6ada17406			
SHA-1:	05c79c4913892219754bb52c62e2a948eb719cfd			

DJI_0018_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	7eecd8b9bdc561442c19cd3b6c462e508398bb945657f0acb4fb3a3b29b18f55			
SHA-1:	47f4b85e8c71ec7a362b0772e9fda9d7f4acd512			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Ios

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	2	2	0
Databases	12	0	0
Drone Logs	1	2	1
Flight Logs	3	0	0
Flight Records	1	1	1
Images	3	3	3
Videos	3	3	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	15	15	0
Videos	10	10	0

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-15 14:50:33 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df005_external.001

2026-06-15 14:50:33 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 8192 df005_external.001

2026-06-15 14:51:39 parser_ios v- rc=n/a

cmd: DF005_flight.zip C:\Users\Floris\Documents\dfdof_output\CASE-df005-2017-ios\p2_image_parsing\controller_ios_parsed
out: controller_ios_parsed

2026-06-15 14:51:53 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797701
out: DJI_0001.MOV

2026-06-15 14:52:05 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797702
out: DJI_0002.JPG

2026-06-15 14:52:06 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797703
out: DJI_0003.JPG

2026-06-15 14:52:06 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797704
out: DJI_0004.JPG

2026-06-15 14:52:18 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797705
out: DJI_0005.MOV

2026-06-15 14:52:31 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797706
out: DJI_0006.JPG

2026-06-15 14:52:31 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797707
out: DJI_0007.JPG

2026-06-15 14:52:31 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797708
out: DJI_0008.JPG

2026-06-15 14:52:48 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797709
out: DJI_0009.MOV

2026-06-15 14:52:59 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797710
out: DJI_0010.JPG

2026-06-15 14:52:59 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797711
out: DJI_0011.JPG

2026-06-15 14:53:12 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797712
out: DJI_0012.MOV

2026-06-15 14:53:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797713

out: DJI_0013.JPG

2026-06-15 14:53:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797714

out: DJI_0014.JPG

2026-06-15 14:53:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797715

out: DJI_0015.JPG

2026-06-15 14:53:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797716

out: DJI_0016.JPG

2026-06-15 14:53:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797717

out: DJI_0017.JPG

2026-06-15 14:53:21 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797718

out: DJI_0018.MOV

2026-06-15 14:53:23 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797719

out: DJI_0019.JPG

2026-06-15 14:53:23 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 797720

out: DJI_0020.JPG

2026-06-15 14:53:24 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 798725

out: DJI_0001_1.MOV

2026-06-15 14:53:27 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 798726

out: DJI_0005_1.MOV

2026-06-15 14:53:29 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 798727

out: DJI_0009_1.MOV

2026-06-15 14:53:30 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 798728

out: DJI_0012_1.MOV

2026-06-15 14:53:31 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df005_external.001 798729

out: DJI_0018_1.MOV

2026-06-15 14:54:19 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe 2017-06-29_14-51-13_FLY005.DAT

out: 2017-06-29_14-51-13_FLY005.csv, 2017-06-29_14-51-13_FLY005.kml

2026-06-15 14:54:20 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2017-06-29_14-52-06_.txt DJIFlightRecord_2017-06-29_14-52-06_.csv

out: DJIFlightRecord_2017-06-29_14-52-06_.csv

2026-06-15 14:54:21 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_14_53_12.thumbnail

out: 2017_06_29_14_53_12_exif.json

2026-06-15 14:54:21 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_14_56_54.thumbnail
out: 2017_06_29_14_56_54_exif.json

2026-06-15 14:54:22 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_15_02_05.thumbnail
out: 2017_06_29_15_02_05_exif.json

2026-06-15 14:54:22 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_14_53_12.mp4
out: 2017_06_29_14_53_12_exif.json

2026-06-15 14:54:22 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_14_56_54.mp4
out: 2017_06_29_14_56_54_exif.json

2026-06-15 14:54:23 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_15_02_05.mp4
out: 2017_06_29_15_02_05_exif.json

2026-06-15 14:54:23 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.JPG
out: DJI_0002_exif.json

2026-06-15 14:54:23 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.JPG
out: DJI_0003_exif.json

2026-06-15 14:54:24 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.JPG
out: DJI_0004_exif.json

2026-06-15 14:54:24 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0006.JPG
out: DJI_0006_exif.json

2026-06-15 14:54:24 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0007.JPG
out: DJI_0007_exif.json

2026-06-15 14:54:25 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0008.JPG
out: DJI_0008_exif.json

2026-06-15 14:54:25 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0010.JPG
out: DJI_0010_exif.json

2026-06-15 14:54:25 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0011.JPG
out: DJI_0011_exif.json

2026-06-15 14:54:26 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0013.JPG
out: DJI_0013_exif.json

2026-06-15 14:54:26 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0014.JPG
out: DJI_0014_exif.json

2026-06-15 14:54:26 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0015.JPG
out: DJI_0015_exif.json

2026-06-15 14:54:27 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0016.JPG
out: DJI_0016_exif.json

2026-06-15 14:54:27 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0017.JPG
out: DJI_0017_exif.json

2026-06-15 14:54:27 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0019.JPG
out: DJI_0019_exif.json

2026-06-15 14:54:27 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0020.JPG
out: DJI_0020_exif.json

2026-06-15 14:54:28 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MOV
out: DJI_0001_exif.json

2026-06-15 14:54:29 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.MOV
out: DJI_0005_exif.json

2026-06-15 14:54:29 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0009.MOV
out: DJI_0009_exif.json

2026-06-15 14:54:30 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0012.MOV
out: DJI_0012_exif.json

2026-06-15 14:54:30 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0018.MOV
out: DJI_0018_exif.json

2026-06-15 14:54:31 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001_1.MOV
out: DJI_0001_1_exif.json

2026-06-15 14:54:31 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005_1.MOV
out: DJI_0005_1_exif.json

2026-06-15 14:54:32 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0009_1.MOV
out: DJI_0009_1_exif.json

2026-06-15 14:54:32 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0012_1.MOV
out: DJI_0012_1_exif.json

2026-06-15 14:54:33 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0018_1.MOV
out: DJI_0018_1_exif.json

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.